Chuanjun “Thompson” Zhou

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Education

University of California, Berkeley Expected May 2027
B.A. Applied Mathematics, College of Letters & Science
B.A. Data Science, College of Computing, Data Science, and Society (CDSS)
Data Science Honors Program, 2026–27 — year-long research thesis (in progress)
GPA 3.722/4.00 | Dean’s List, Fall 2023 and Spring 2024

Graduate coursework: Stat 238 Bayesian Statistics (**A+**) • Stat 210A Theoretical Statistics (IP)
Notable coursework: EECS 127 Optimization Models (A) • CS 189 Machine Learning (A–) • CS 188 Artificial Intelligence (A-) • EECS 183 Natural Language Processing (IP) • CS 182 Deep Learning (IP) • CPH 100 Foundations of Computational Precision Health (IP) • Data 144 Data Mining (A) • Data 100 Foundations of Data Science • CS 61B Data Structures • Stat 134 Probability (A) • Stat 150 Stochastic Processes • Math 104 Real Analysis • Math 110 Abstract Linear Algebra • Math 128A Numerical Analysis (A)

Publications and Preprints

1. **C. Zhou** and A. Anderson. “What Transfers Across Cuneiform?” *Proceedings of EMNLP 2026* (main conference, **oral** presentation).
2. H. Gu, **C. Zhou**, et al. “Noise-Aware Report-Supervised Learning: Decomposing Report-Quality and Extraction Noise.” *In preparation; target CVPR 2027.*
3. **C. Zhou**, A. Anderson. “OWLID-OCR: Whose Word Is This? Open-World Identification of Ancient Languages Embedded in Modern Scholarship.” *In preparation; target ACL 2027.*

Research Experience

Machine Learning Research Intern — UCSF Radiology and Biomedical Imaging May 2026 – present
UCSF *Prof. Yang Yang, Prof. Kang Wang; Dr. Hanxue Gu*

Report-Supervised Tumor Segmentation — learning to segment tumors from radiology reports when hand-drawn masks are scarce, and deciding how much each report statement should be trusted.

- **Leading the implementation and experiments** for our CVPR 2027 submission on abdominal CT (liver, kidney, pancreas). Built the full pipeline on UCSF’s GPU cluster: LLM parsing of 2,760 radiology reports into 7,442 tumor-level records, organ masks for 2,764 CT scans, and a verified reproduction of R-Super training.
- Implemented calibrated report supervision inside the R-Super loss (my honors thesis): separate trust for each kind of report claim, such as a lesion’s size, a hedged finding, or a “no tumor” statement, fitted on a small set of masked cases and frozen before training. Each constraint is blended between strict and relaxed versions by that trust, with unit tests and pre-registered experiments.
- Found that the released R-Super loader **trains on only about a quarter of report-labeled scans**, because it requires a numeric size for every lesion; our version supervises from all usable reports.
- Earlier, on prostate MRI: built a 12-field LLM extraction schema over 1,471 reports, ported R-Super from CT to MRI, and trained MedFormer (test Dice 0.70). Found that **the objective reads only lesion size and location**, so most extracted fields never reach the model, and that unreported lesions raise false positives that best-threshold F1 hides.

Undergraduate Researcher, Computational Linguistics and NLP — Fact Grid Cuneiform Jan 2026 – present
UC Berkeley *Collaborator: Prof. Adam Anderson*

Cross-lingual transfer and word segmentation for Akkadian, Sumerian, Elamite, and Hittite.

- **First author** on "What Transfers Across Cuneiform?" (EMNLP 2026, main conference). Built the benchmark and the cunei-tools library and established the central finding: **lexicon coverage of the target language predicts Morfessor's transfer at r = 0.91** across 140 runs (14 source sizes × 5 seeds), while transitional-probability segmentation holds F1 > 0.95 across all six language pairs. Showed the coverage effect is causal, not correlational: clamping a high-resource source to a low-resource target's sign inventory collapses transfer (0.857 → 0.283), and matched-size controls separate the effect of coverage from the effect of corpus size.
- **Leading the modeling for OWLID-OCR** (in preparation; target ACL 2027), an OCR post-correction system for cuneiform: a noisy-channel generative formulation with an open-set unknown-token mechanism and hierarchical pooling across small embedded-language lexica.
- Built an end-to-end pipeline converting noisy transliterations to normalized cuneiform (95%+ conversion on ~2M tokens); fine-tuned a BART seq2seq model for lemmatization and morphological tagging.

Teaching

Academic Student Employee (Tutor) — UC Berkeley CDSS 3 semesters

- **Data C88C**, Computational Structures in Data Science Aug 2025 – Jun 2026
- **Data C88S**, Probability and Mathematical Statistics in Data Science Jan – May 2025

Ran office hours and small-group instruction, graded and identified common error patterns, **authored midterm and final examination problems**, and maintained the course repository and Ed forum.

Technical Skills

Languages: Python, R, SQL, Java, MATLAB

Machine learning: PyTorch, HuggingFace Transformers, scikit-learn, NumPy/SciPy/pandas/Polars

HPC: SLURM, distributed and mixed-precision training, offline model serving on air-gapped clusters

Research areas: 3D medical image segmentation, weak and noisy supervision, low-resource NLP, Bayesian modeling, convex optimization

Human languages: English (native), Mandarin (native)